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Department of Computer Science
Academic Year: 2024-25

F.Y.B.Sc.(Data Science) Sem – I
Problem Solving and Python Programming
Practical Assignment6

Date: 27/09/2024

1. Write a Python program to define a Wheel class with attributes size and type. Implement a method describe_wheels() that prints out the details of each wheel.
2. Define a Car class include a constructor that initializes make and model. Add an instance method start_engine() that prints a message "Engine started for {make} {model}".
3. Create a class ElectricCar that inherits from Car. Add an additional attribute range_per_charge and a method display_range() that prints the range per charge.
4. Write a Python program to create a person class. Include attributes like name, country and date of birth. Implement a method to determine the person's age.
5. Design a class structure for a simple banking system. Create classes for BankAccount, SavingsAccount, and CheckingAccount. Ensure appropriate inheritance and include methods for deposit, withdraw, and check balance.
6. Create a base class Shape with a method area() that returns 0. Create derived classes Rectangle and Circle that inherit from Shape and override the area() method to calculate the area of a rectangle and circle, respectively.

PROGRAMS FOR PRACTICE:

7. Write a Python program to create two empty classes, Student and Marks. Now create some instances and check whether they are instances of the said classes or not. Also, check whether the said classes are subclasses of the built-in object class or not.
8. Define classes A, B, and C where B and C inherit from A. Implement a method method_A() in A, and methods method_B() and method_C() in B and C respectively. Create a class D that inherits from both B and C and calls all three methods.
9. Define a base class Vehicle with a method drive() that prints "Driving a vehicle". Create derived classes Car and Motorcycle that inherit from Vehicle. Override the drive() method in Car to print "Driving a car" and in Motorcycle to print "Riding a motorcycle".
10. Define a base class Parent with a parameterized constructor that initializes a name attribute. Create a derived class Child that inherits from Parent and initializes both name and age attributes using the constructor of the parent class.
11. Define a class Shape with a method draw() that prints "Drawing a shape". Create derived classes Rectangle and Circle that override draw() to print "Drawing a rectangle" and "Drawing a circle", respectively. Demonstrate polymorphism by calling draw() on instances of both classes.
12. Define a class Vector that takes two coordinates (x and y) and implements the + operator to add two vectors, and * operator for scalar multiplication. Test these operators with instances of Vector.

13. Create a class `Employee` with a method `calculate_salary()` that returns a base salary. Define subclasses `Manager` and `Developer` that override `calculate_salary()` to include bonuses. Demonstrate polymorphism by calling `calculate_salary()` on instances of both subclasses.
14. Write a Python class `Inventory` with attributes like `item_id`, `item_name`, `stock_count`, and `price`, and methods like `add_item`, `update_item`, and `check_item_details`